

Extending the Service Life of High-Ni Layered Oxides by Tuning the Electrode–Electrolyte Interphase

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As a high-energy-density cathode for Li-ion batteries, high-Ni layered oxides, especially with ultrahigh Ni-content, suffer from short lifespans, due in part to their unstable electrode–electrolyte interphase (EEI). Herein, the cycle life of $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ is greatly extended by manipulating the EEI with a lithium bis(oxalate) (LiBOB) additive even when operated at a moderately high voltage (4.4 V vs Li/Li^+). Impressively, the capacity retention can be increased from 61 to 80% after 500 cycles in a full cell paired with a graphite anode. Additionally, the presence of LiBOB enables a robust boron- and oxygen-enriched EEI that effectively inhibits continual electrolyte decomposition and offers a stable cathode surface. Moreover, the layered architecture of the cathode–electrolyte interphase (CEI) and the anode–electrolyte interphase (AEI) at the nanometer scale is revealed by time-of-flight secondary ion mass spectrometry. It is demonstrated that the cathode surface chemistry can significantly influence the AEI both chemically and physically, and AEI is modified from a thick “three-layer” to a thin “two-layer” architecture by tuning the cathode surface chemistry with LiBOB. This work presents a correlation between the EEI characteristics and battery performance and highlights the significance of manipulating surface chemistry in developing stable high-energy-density Li-ion batteries.

1. Introduction

The rapidly growing energy market calls for advanced Li-ion batteries with high energy density and acceptable service life. Ni-rich layered oxides ($\text{LiNi}_x\text{M}_{1-x}\text{O}_2$, $x > 0.6$, $\text{M} = \text{Mn, Ni, Co, Al}$), especially those with ultrahigh Ni content ($\text{Ni} > 0.9$), are able to deliver a high capacity of $>200 \text{ mA h g}^{-1}$ with a high average voltage (3.8 V vs Li/Li^+). They hold great promise to power a 300-mile electric vehicle and are being intensively studied in recent years.^[1] However, degradation of the electrode–electrolyte interphase, which is a major cause of poor cycle life, remains as a critical problem that hinders their widespread application, especially when the voltage exceeds the HOMO (highest occupied molecular orbital, $\approx 4.3 \text{ V vs Li/Li}^+$) of the

organic carbonate electrolytes used.^[2–5] Specifically, the parasitic electrolyte decomposition occurs on the charged cathode surface, resulting in the formation of cathode–electrolyte interphase (CEI) with complicated surface chemistry.^[6–11] Moreover, the surface degradation on cathode usually involves active mass dissolution, which is a common phenomenon in various cathodes.^[12] Subsequently, the dissolved transition-metal ions reach the anode side through chemical crossover and poison the anode–electrolyte interphase (AEI) on graphite surface.^[13,14] Thus, understanding the chemical and electrochemical reaction between the electrode and electrolyte, which influences the composition, microstructure, and chemical properties of the formed electrolyte–electrolyte interphases (EEIs), is crucial to establishing high-energy-density Li-ion batteries with long, stable service life.

Conventionally, inert materials coating (e.g., Al_2O_3 ,^[15] AlPO_4 ,^[16] AlF_3 ,^[17] etc.) has been applied to modify the cathode sur-

face chemistry and mitigate the interphase degradation on the cathode. Nevertheless, the cathode bulk particle could gradually lose contact with the coating layer due to the anisotropic volume change during lithiation and delithiation.^[18,19] On the other hand, electrolyte additive, which is able to regulate the EEI chemistry through an in situ modification strategy, has been widely studied and proved to ensure high Coulombic efficiency and voltage efficiencies with long cycle life.^[20–23] However, due to the complexity and air-sensitivity of the EEI chemistry, there is a limited understanding of the EEI configuration and architecture, and how different components influence the EEI layer properties and battery performance is yet to be fully understood.^[24–26]

Herein, with an ultrahigh-Ni layered oxide ($\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$) as a model cathode and lithium bis(oxalate) borate (LiBOB) as a model electrolyte additive, the EEI chemical and physical property changes on both the cathode and anode are elucidated. Moreover, the layered architecture of the CEI and AEI at the nanometer scale is revealed by time-of-flight secondary ion mass spectrometry (TOF-SIMS) and the correlation between CEI and AEI properties is illustrated. On a chemical perspective, the tuned EEIs are configured with B_xO_y species and less ethylene carbonate/LiPF₆ decomposition products, which endows the CEI with extreme robustness against

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electrochemical abuse and better Li-ion diffusivity within both the CEI and AEI. Moreover, it reveals that the modification of the surface chemistry on the cathode side would beneficially change the physical stacking behavior of components on the AEI. AEI is reconstructed from a thick “three-layer” architecture (F-enriched exterior layer, O-dominating intermediate layer, and Li-containing interior layer) to a thin “two-layer” architecture (O-enriched exterior layer and Li-dominating interior layer) benefiting from the B_xO_y enrichment within the EEI. As a result, superior electrochemical performance is achieved by manipulating the electrode surface chemistry even under a moderate high operating voltage (4.4 V vs Li/Li^+), and the capacity retention is greatly increased from 61 to 80% after 500 cycles in a full cell assembled with graphite anode.

2. Results and Discussion

2.1. Formation of LiBOB-Tuned EEI

Figure 1a depicts the initial charge–discharge curves of $Li | LiNi_{0.94}Co_{0.06}O_2$ half cells with various electrolytes under a current rate of C/10 (more repeated data are shown in Figure S1, Supporting Information). The initial charge and discharge

capacities of the baseline cell without LiBOB additive (denoted as “baseline”) are, respectively, 258 and 228 $mA h g^{-1}$, corresponding to a Coulombic efficiency of 88.2%. As for the cells with 0.3 wt% LiBOB-added electrolyte (denoted hereafter as “0.3% LiBOB”) and 1.5 wt% LiBOB-added electrolyte (denoted hereafter as “1.5% LiBOB”), the charge capacities are increased marginally to, respectively, 260 and 263 $mA h g^{-1}$, with Coulombic efficiencies of 86.8 and 85.6%. The lower Coulombic efficiency with higher LiBOB content in the electrolyte is mainly due to the sacrificial oxidation of BOB^- anion group at high potential (Figure 1a inset), which agrees well with previous studies.^[27,28] Additionally, Figure 1b displays a 3D view of the BO^- fragment distribution in the TOF-SIMS sputtered volume collected on the cycled $LiNi_{0.94}Co_{0.06}O_2$ surface (retrieved from full cells paired with a graphite anode after 500 cycles) with and without LiBOB. Compared with the baseline cathode sample, the BO^- signal on the 1.5% LiBOB sample is significantly stronger and shows an almost uniform distribution in CEI. Thus, the CEI composition is substantially modified by the oxidative decomposition of LiBOB during initial cycling.

On the other hand, the AEI chemistry on the graphite anode is also tuned by LiBOB. Figure 1c shows the initial charge curves of graphite $| LiNi_{0.94}Co_{0.06}O_2$ full cells with various electrolytes. With 1.5% LiBOB, a new voltage plateau appears at

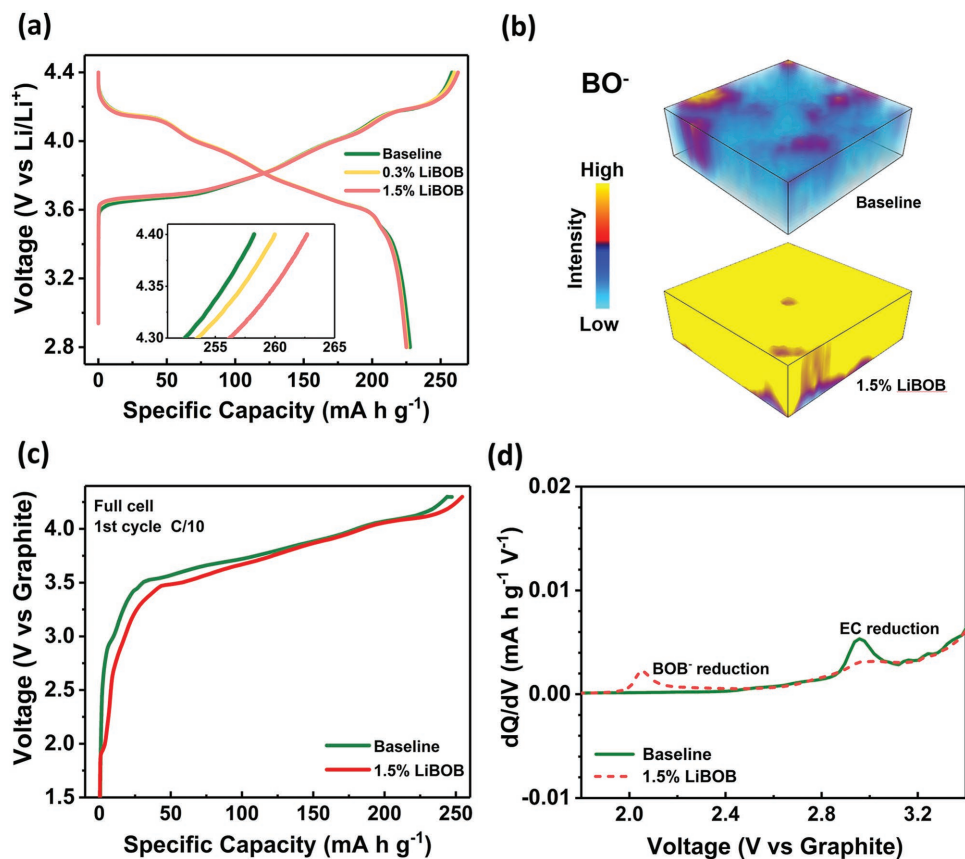


Figure 1. a) The initial charge–discharge curves of $Li | LiNi_{0.94}Co_{0.06}O_2$ half cells with different electrolytes. b) 3D distribution of LiBOB decomposition species (represented by BO^-) in the TOF-SIMS sputtered volume on $LiNi_{0.94}Co_{0.06}O_2$ cathode surface retrieved from 500-cycle full cells with baseline electrolyte and 1.5% LiBOB. Intensity of the BO^- fragment is normalized to maximum in the baseline electrolyte, and size of the TOF-SIMS sputtered volume is 100 μm (length) $\times$ 100 μm (width) $\times$ 22 nm (height). c) The initial charge voltage profiles and d) corresponding dQ/dV curves of graphite $| LiNi_{0.94}Co_{0.06}O_2$ full cells with baseline electrolyte and 1.5% LiBOB.

≈ 2.0 V without observable plateaus at 3.0 V as compared to the baseline electrolyte. Furthermore, the corresponding differential capacity (dQ/dV) profile (Figure 1d) provides a more detailed and straightforward comparison. The full cell with the baseline electrolyte has a large peak at 3.0 V, resulting from the reduction of ethylene carbonate (EC) in forming AEI on the graphite surface.^[29] In contrast, the full cell with the LiBOB-added electrolyte shows a peak at a lower potential of 2.1 V, which is due to the reductive decomposition of BOB[−] oxalate ligand prior to EC reduction.^[30,31] Meanwhile, the declined peak intensity at 3.0 V indicates significantly suppressed contribution of EC decomposition to the AEI formation. Therefore, the AEI chemical property on the graphite surface is effectively modified by the sacrificial reduction of the BOB[−] oxalate ligand.

2.2. Electrochemical Performance

The electrochemical performance of Li | LiNi_{0.94}Co_{0.06}O₂ half cells with different electrolytes is summarized in Figure 2. Figure 2a displays the discharge capacities of the half cells with baseline electrolyte and 0.3 and 1.5% LiBOB-added electrolytes at various current rates. Initially, at C/3 rate, all cells go through an activation process with increasing discharge capacities upon cycling. Notably, owing to the ultrahigh Ni-content in LiNi_{0.94}Co_{0.06}O₂, the output discharge capacities of all these cells

reach over 220 mA h g^{−1}, which are among the highest values reported for the high-Ni layered oxide system.^[32–34] Furthermore, it reveals that the cells with LiBOB exhibit comparatively better rate capability, and this trend is more pronounced as the current rate increases. For example, at 9C rate, the baseline cell delivers a discharge capacity of 179 mA h g^{−1}, in comparison to the higher capacities of 182 and 185 mA h g^{−1} obtained in cells with, respectively, 0.3 and 1.5% LiBOB. The improved rate performance with LiBOB addition can be ascribed to the modified cathode surface chemistry with higher Li-ion diffusivity, which will be discussed later with the X-ray photoelectron spectroscopy (XPS) and TOF-SIMS analysis. Note the concern of excessive decomposition of LiBOB at extreme potentials (e.g., 5.0 V vs Li/Li⁺) has been reported to cause overthickening of CEI and large impedance buildup.^[35–37] Here, with a comparatively lower upper cutoff voltage (4.4 V vs Li/Li⁺), the LiBOB-tuned CEI can be well tailored without overthickening, which is critical to the superior rate performance.

Moreover, Figure 2b illustrates the cycling performance of half cells with varied electrolytes at C/3 rate. During the formation cycles at C/10 rate, all cells could approach superb discharge capacities of ≈ 230 mA h g^{−1}. As cycling proceeds, however, the cells with LiBOB achieve much better capacity retention and higher Coulombic efficiency (Figure S2, Supporting Information). Specifically, the baseline cell only attains a capacity retention of 82%, while higher capacity retentions of 88 and 92% are

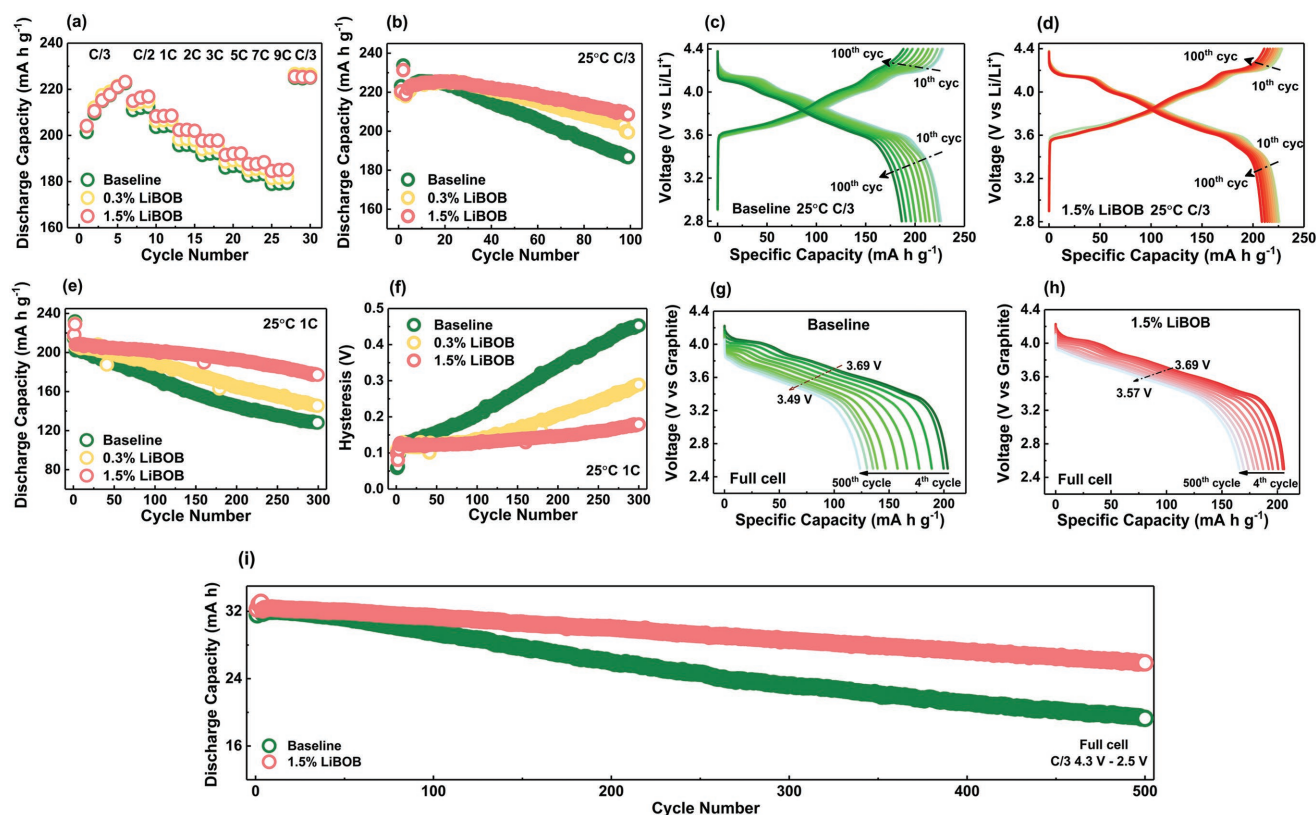


Figure 2. Electrochemical characterization of Li | LiNi_{0.94}Co_{0.06}O₂ half cells with different electrolytes: a) rate capability, b) capacity retention at C/3 rate, and c,d) corresponding charge–discharge curves of various cycles, e) cycling performance at 1C rate, and f) hysteresis evolution at 1C rate. The voltage window for half cells is 4.4–2.8 V versus Li/Li⁺. Electrochemical characterization of graphite | LiNi_{0.94}Co_{0.06}O₂ full cells: discharge curves with g) baseline electrolyte and h) 1.5% LiBOB, and i) corresponding cycling performance at C/3 rate. 1C = 180 mA g^{−1} for both half and full cells.

achieved in cells with, respectively, 0.3 and 1.5% LiBOB. Moreover, Figure 2c,d further compares the corresponding charge–discharge profiles of the baseline and 1.5% LiBOB cells from the 10th to the 100th cycle. At the tenth cycle, both cells demonstrate similar voltage curves with two distinguishable plateaus at around 4.0 and 4.2 V, corresponding, respectively, to the hexagonal 1 to hexagonal 2 (H1–H2) and hexagonal 2 to hexagonal 3 (H2–H3) phase transition processes.^[38,39] After 100 cycles, for the baseline cell, both the previously observed voltage plateaus disappear, implying a severely distorted crystal structure with deteriorated lithiation/delithiation kinetics. In sharp contrast, the cell with 1.5% LiBOB still preserves well-established plateaus. Furthermore, Figure 2e depicts the cycling performance of cells with different electrolytes at 1C rate. After 300 cycles, the cell with 1.5% LiBOB shows a capacity retention of 85%, which is much higher than that with 0.3% LiBOB and baseline electrolyte (capacity retention of 70 and 62%, respectively). Additionally, the hysteresis (defined as the difference between the average charge and discharge voltages) is greatly reduced to 0.29 and 0.18 V for the cells with, respectively, 0.3% LiBOB and 1.5% LiBOB as compared to the high hysteresis value of 0.45 V in the baseline cell (Figure 2f). Furthermore, the Coulombic efficiency increases as the concentration of LiBOB in the electrolyte increases (Figure S3, Supporting Information). However, considering that the lithium metal could also influence the performance of half cells, a galvanostatic Li plating/stripping electrochemical cycling measurement was performed to investigate the dynamical stability of the Li/electrolyte interface to rule out the influence of LiBOB on the lithium metal. As shown in Figures S4–S6 (Supporting Information), the lithium stripping and plating behavior is not influenced by the addition of LiBOB, and the Li|Li symmetric cells exhibit an overpotential of 0.10, 0.10, and 0.12 V when cycled with, respectively, baseline electrolyte, 0.3% LiBOB, and 1.5% LiBOB. Additionally, the lithium metal retrieved from half cells after 300 cycles exhibits more mossy structures with the addition of LiBOB as evidenced in Figure S7 (Supporting Information). Thus, it is reasonable to conclude that the prominent enhancement in capacity and working voltage retention brought by LiBOB addition mainly results from the formation of a robust protective CEI on $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ tuned by the oxidative decomposition of the BOB^- ligand during cycling (as confirmed by the detailed analyses of XPS and TOF-SIMS shown later), which inhibits unwanted parasitic electrode–electrolyte reactions and enables superior cell reaction kinetics.

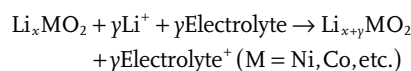
From the half-cell test results discussed above, it is demonstrated that the addition of 1.5% LiBOB to the electrolyte maximizes the cyclability of $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$. For practical applications, the robustness of LiBOB-tuned EEI is also evaluated in pouch-type full cells employing the $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ cathode, graphite anode, and different electrolytes (baseline and 1.5% LiBOB) for long-term cycling. As shown in Figure 2i, the full cell with 1.5% LiBOB could achieve a capacity retention of as high as 80% after 500 cycles. In contrast, the baseline cell suffers from a much more severe capacity decay, retaining only 61% of its original capacity. Moreover, with 1.5% LiBOB addition, the voltage drop on discharge process decreases from 0.20 to 0.12 V (dashed lines in Figure 2g,h) and the Coulombic efficiency increases from ≈ 99.0 to $>99.5\%$ (Figure S8,

Supporting Information). These results demonstrate that the long-term cyclability of $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ with ultrahigh Ni-content can be successfully achieved in practical cell configurations with the well-tailored EEI contributed by the LiBOB incorporation in the electrolyte.

2.3. Surface Analyses

2.3.1. Cathode Surface Stability

To investigate the origin of the significantly improved electrochemical performance of $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ in both half and full cells with LiBOB addition in the electrolyte, self-discharge experiment was carried out to evaluate the cathode surface stability at high voltage. The self-discharge of Ni-rich layered oxides at charged states follows the reaction^[27]



In this work, the $\text{Li}|\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ cells with different electrolytes were charged to 4.4 V versus Li/Li^+ at C/3 rate and then rested for 72 h under 55 °C, followed by discharging to 2.8 V. As a reference, the cell employing baseline electrolyte without resting after charging to 4.4 V was also tested. During charge (Figure 3a), the cell with 1.5% LiBOB delivers a higher charge capacity than the baseline cell at 55 °C, which can be attributed to the BOB^- oxalate ligand oxidation at elevated temperature. Figure 3b illustrates variations on the open-circuit voltage (OCV) during subsequent resting. The baseline cell suffers from a rapid OCV drop and ends up with 4.02 V, implying fast self-discharge (relithiation) during rest. After rest, it only delivers a discharge capacity of 157 mA h g^{−1} (Figure 3c), much lower than that of the baseline cell without resting at 4.4 V (242 mA h g^{−1}, dashed line in Figure 3c). On the contrary, the cell with 1.5% LiBOB exhibits higher resistance against self-discharge, with the OCV value remaining at 4.16 V and a remarkable discharge capacity of 220 mA h g^{−1} after 72 h of storage. These results clearly evidence the protective role of LiBOB-tuned CEI on suppressing unwanted side reactions between the high-Ni layered oxide cathode and electrolyte at high voltages.

Moreover, Ni-rich layered oxides tend to undergo a structural degradation from the *R-3m* phase to a rock salt NiO phase (*Fm-3m*) during repeated lithiation/delithiation, which could be accelerated by aggressive electrolyte attack at the particle surface followed by gradual propagation to the bulk.^[8,40] The degree of the rock salt phase formation can be evaluated by the intensity ratio of (003) versus (104) peaks through X-ray diffraction (XRD). The higher the $I(003)/I(104)$ ratio, the less the rock salt phase is formed.^[41] Figure 3d highlights the effect of the LiBOB-tuned CEI on preserving structural stability of $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ during cycling. Notably, the $I(003)/I(104)$ ratio is much higher in $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ with 1.5% LiBOB (1.16) than that in the baseline cathode (1.04) after 500 cycles in full cells, indicative of the enhanced structural reversibility with LiBOB during long-term cycling. Furthermore, the separation between (006) and (102) peaks vanishes in the baseline cathode, suggesting

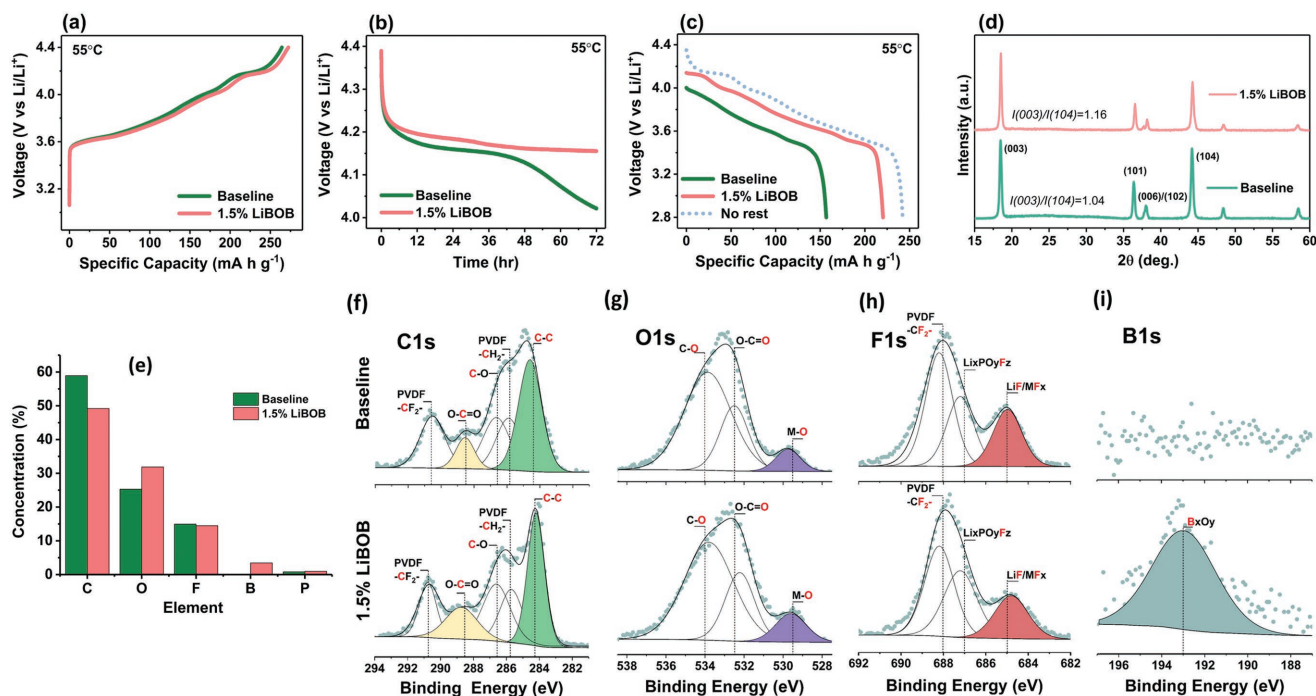


Figure 3. Self-discharge tests of $\text{Li}|\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ half cells at 55 °C with baseline electrolyte and 1.5% LiBOB: a) charge voltage curves, b) OCV evolution during resting for 72 h, and c) subsequent discharge curves after resting. d) XRD patterns of $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ cathodes harvested from graphite $|\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ full cells after 500 cycles. XPS characterization of the $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ cathodes from $\text{Li}|\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ half cells after 100 cycles with baseline electrolyte and 1.5% LiBOB: e) elemental concentration, f) C1s, g) O1s, h) F1s, and i) B1s spectra.

the severe structural deterioration. For $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ with 1.5% LiBOB, however, this peak separation is well maintained and distinguishable, confirming the improved structural integrity after prolonged cycling.^[17] Thus, by alleviating parasitic cathode–electrolyte reactions with LiBOB-tuned CEI, the structural integrity of $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ can be well preserved with inhibited detrimental phase transformation.^[42]

2.3.2. Cathode Surface Chemistry

To elucidate the effect of LiBOB on the CEI chemical composition, XPS data were collected on cycled $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ cathodes and graphite anodes. Figure 3e compares the CEI elemental composition with various electrolytes. With 1.5% LiBOB, the concentration of B species increases from none to 3.5 at%, while the C species decreases from 58.9 to 49.2 at%. In the meantime, the O content increases from 25.3 to 31.9 at% and the F/P content remains almost identical. This comparison indicates that the LiBOB-tuned CEI is boron- and oxygen-rich, which originates from the participation of BOB^- ligand in the CEI formation upon initial cycling. High-resolution XPS spectra in Figure 3f–i offer deeper insight into the CEI composition. To begin with, for $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ with 1.5% LiBOB, the concentration of C–C bonding (284.3 eV, originating from carbon black or chemisorbed adventitious carbon) decreases while the $-\text{CO}_2$ concentration (288.5 eV) increases compared to the baseline cathode (Figure 3f). The higher $-\text{CO}_2$ content is consistent with previous studies revealing that the BOB^-

ligand contributes to more lithium oxalates and ROCO_2Li formation.^[43,44] Second, in Figure 3g, the content of O–M bonding (529.5 eV), which arises from the active bulk material and could serve as an indicator for the CEI thickness, is similar in both samples. This suggests that the CEI thickness is almost unchanged with LiBOB incorporation, in good line with the moderate oxidative decomposition of the BOB^- oxalate ligand at 4.4 V versus Li/Li^+ as indicated before. Moreover, with 1.5% LiBOB, there is a decrease in the concentration of electronically insulating LiF/MF_x species (684.5–685.5 eV) on CEI (Figure 3h), which mainly stems from the LiPF_6 decomposition products or dissolved transition-metal cations aggravated by acidic species attack (e.g., HF and PF_5 from LiPF_6 hydrolysis). Thus, the reduced LiF/MF_x content suggests alleviated attack from the electrolyte on the cathode surface with the LiBOB-tuned CEI. Finally, a strong B–O peak emerges at 193.0 eV referring to B_xO_y species (Figure 3i),^[45] which further confirms the contribution of BOB^- oxalate ligand to the CEI formation. Together, these data demonstrate that LiBOB is preferentially decomposed at the surface of $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ to form a boron- and oxygen-enriched protective CEI, which is the origin for the improved ionic diffusivity and superior electrochemical stability at high voltages.

Insights into the CEI configuration at the nanometer level are further provided by TOF-SIMS—a highly surface-sensitive technique with ultrahigh chemical selectivity.^[46] Figure 4a–c shows the distribution profiles of the selected secondary-ion fragments acquired from the cycled $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ with different electrolytes, including C_2HO^- (representing organic

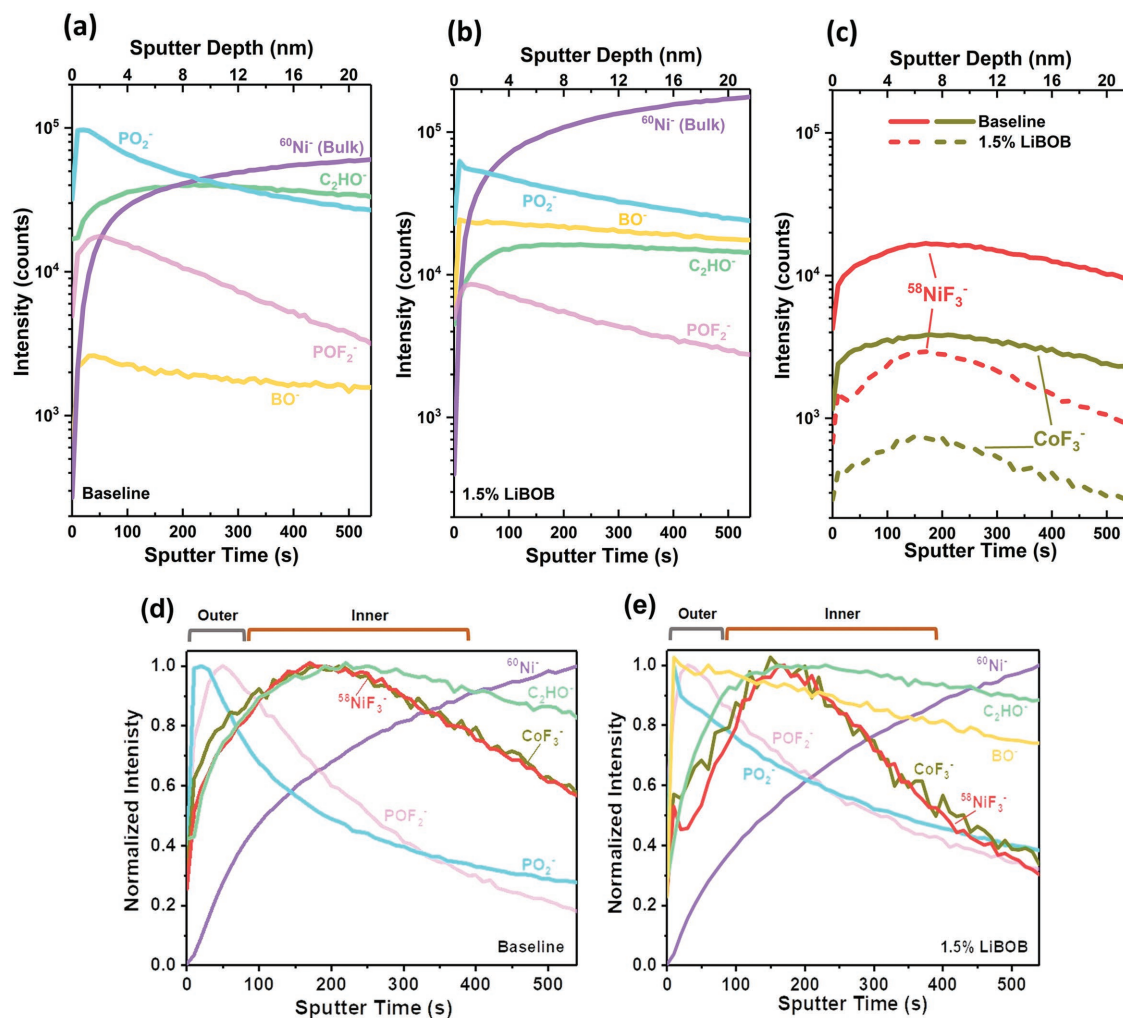


Figure 4. TOF-SIMS depth profiles of fragments of interest acquired on $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ cathodes retrieved from 500-cycle full cells with a) baseline electrolyte and b) 1.5% LiBOB, including $^{60}\text{Ni}^-$ (representing bulk $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$), $\text{PO}_2^-/\text{POF}_2^-$ (representing inorganic CEI species related to LiPF_6 decomposition), C_2HO^- (representing organic CEI species from carbonate solvent decomposition), and BO^- (representing CEI species from LiBOB decomposition). c) Comparison of TOF-SIMS depth profiles on transition-metal dissolution (represented by $^{58}\text{NiF}_3^-$ and CoF_3^-) and CEI architecture (represented by the above fragments of interest; normalized to maximum) d) without and e) with LiBOB.

interphasial species from the decomposition of carbonate electrolyte),^[47,48] $\text{PO}_2^-/\text{POF}_2^-$ (representing inorganic interphasial species related to LiPF_6 decomposition),^[49] $^{60}\text{Ni}^-$ (representing bulk $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$), $^{58}\text{NiF}_3^-/\text{CoF}_3^-$ (representing active mass dissolution products),^[13] and BO^- (representing interphasial species from LiBOB decomposition). Specifically, for $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ with 1.5% LiBOB, the amount of B-bearing components (represented by BO^-) increases significantly within CEI. On the other hand, the intensities of other common interphasial species including C_2HO^- , PO_2^- , and POF_2^- decline within CEI with 1.5% LiBOB. Meanwhile, active mass (transition-metal) dissolution is effectively reduced, as evidenced by much weaker $^{58}\text{NiF}_3^-/\text{CoF}_3^-$ signals within the CEI (Figure 4c) and their shorter tails after reaching maximum intensities at a sputtering depth of ≈ 7 nm (175 s of Cs^+ sputtering, Figure 4d,e) from the top surface of CEI. These results are in good agreement with the above-discussed self-discharge and XPS data, further corroborating the protective role of LiBOB-tuned

CEI in $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ that considerably alleviates undesired electrode–electrolyte reactions.

It should be noted that there is virtually no difference on the CEI architecture in $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ with baseline electrolyte and 1.5% LiBOB. As shown in Figure 4d,e, CEIs for both samples exhibit a “two-layer” structure, with the exterior layer majorly comprising P-bearing species (e.g., PO_2^- , POF_2^-) while the interior layer enriched with organic species (C_2HO^-). However, the distribution behavior of the dissolved transition-metal species ($^{58}\text{NiF}_3^-$, CoF_3^-) is distinct. The intensities of $^{58}\text{NiF}_3^-$ and CoF_3^- fragments for both samples reach their peak values at a similar depth of ≈ 7 nm (175 s of sputter time) from the CEI surface. As the sputtering goes on, the signals quickly decline and end up with 30% of the peak intensity in $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ with 1.5% LiBOB. However, this decline trend is retarded for the baseline sample and the intensity remains at 55% of peak intensity at the end of sputtering. It is a good evidence showing the excellent acidic resistance of the boron and oxygen-enriched

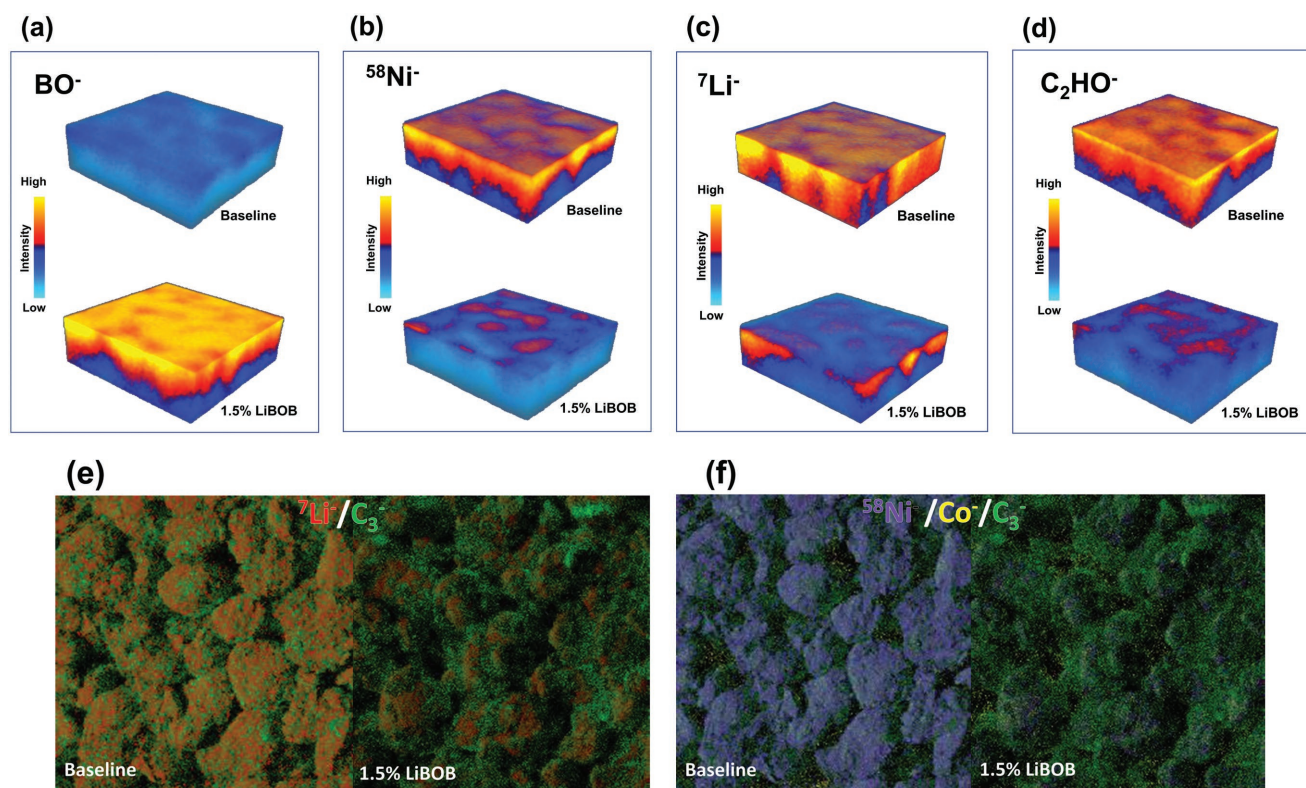


Figure 5. 3D distributions of a) LiBOB decomposition species (represented by BO^-), b) deposited transition-metal species (represented by $^{58}\text{Ni}^-$), c) trapped lithium (“dead” Li; represented by $^7\text{Li}^-$), and d) carbonate solvent decomposition species (represented by C_2HO^-) in TOF-SIMS sputtered volume, collected on graphite anodes retrieved from 500-cycle full cells with baseline electrolyte and 1.5% LiBOB. Note that intensity of the fragments is not normalized. Size of the TOF-SIMS sputtered volume is $100\text{ }\mu\text{m}$ (length) $\times$ $100\text{ }\mu\text{m}$ (width) $\times$ 17 nm (height). TOF-SIMS chemical maps of e) $^7\text{Li}^-/\text{C}_3^-$ and f) $^{58}\text{Ni}^-/\text{Co}^-/\text{C}_3^-$ after Cs^+ sputtering for 150 s acquired on graphite harvested from 500-cycle full cells with baseline electrolyte and 1.5% LiBOB.

CEI, which prevents continuous active mass dissolution and maintains structural integrity, in accord with the previous discussion about cathode surface stability.

2.3.3. Anode Surface Chemistry

The AEI chemical composition at the graphite anode surface is first analyzed by XPS. Figure S11a (Supporting Information) depicts the elemental concentration of the graphite AEI with various electrolytes. Similar to the cathode side, the graphite surface manifests a boron- and oxygen-enriched AEI with 1.5% LiBOB, as evidenced by the enhancement of O and B contents, respectively, from 21.0 to 28.5 at% and none to 4.3 at% due to the participation of LiBOB. Meanwhile, although the C content appears roughly constant, the high-resolution C1s spectra (Figure S11b, Supporting Information) provide more specific information about the compositional discrepancy between these two samples. Specifically, with 1.5% LiBOB, the contents of C–C bonding (284.3 eV) and $-\text{CO}_3$ bonding (290.0 eV) are much higher compared to that with baseline electrolyte. And the increased content of $-\text{CO}_3$ bonding is mainly ascribed to BO^- ligand decomposition.^[43] As for the C–C bonding, given that it primarily arises from the bulk graphite, the content of C–C bonding could act as an indicator to evaluate the AEI thickness, meaning the higher the C–C content, the thinner

the AEI formed on the graphite surface. Based on the higher content of the C–C bonding, the graphite AEI becomes thinner with 1.5% LiBOB. Another significant difference is the Li content, which decreases from 24.1 to 14.8 at% after adding 1.5% LiBOB to the electrolyte. Note that the trapped Li on the graphite surface, conventionally known as “dead” Li, can originate from deposited metallic Li microstructures (e.g., dendrites) during long-term cycling and could pose severe safety concerns.^[50–54] Interestingly, despite contributions from LiBOB decomposition, the Li content is much less in the LiBOB-tuned AEI with respect to the baseline sample after 500 cycles. In addition, much less electronically insulating LiF/MF_x species are revealed at the graphite surface with LiBOB addition in the high-resolution F1s spectra (Figure S11d, Supporting Information), suggesting greatly suppressed transition-metal dissolution from cathode.

The composition of graphite AEI was further investigated with TOF-SIMS as shown in Figure 5a–d. To begin with, with 1.5% LiBOB incorporation, the exterior AEI is enriched with B-bearing species (represented by BO^- in Figure 5a and BO^- in Figure S15, Supporting Information), corresponding to B_xO_y species in Figure S11e (Supporting Information). Owing to a robust LiBOB-tuned CEI on $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$, which effectively suppresses active mass dissolution and crossover (Figure 4c), the deposited transition-metal species at the graphite AEI (represented by $^{58}\text{Ni}^-$ in Figure 5b with quantitative comparison

in Figure S16a and Table S2, Supporting Information, as well as $^{58}\text{Ni}^-$, $^{58}\text{NiF}^-$, $^{58}\text{NiO}^-$ in Figure S16b, Supporting Information) are substantially reduced. Meanwhile, the amount of Li-containing species (represented by $^7\text{Li}^-$ in Figure 5c with quantitative comparison in Figure S16a and Table S3, Supporting Information, as well as $^7\text{LiF}_2^-$, $^7\text{Li}^-$, and $^7\text{LiO}^-$ in Figure S16c, Supporting Information) considerably decreases as well. These observations are further supported by the high-resolution TOF-SIMS chemical maps of $^7\text{Li}^-/\text{C}_3^-$ and $^{58}\text{Ni}^-/\text{Co}^-/\text{C}_3^-$ fragments on graphite AEI after 150 s of Cs^+ sputtering in Figure 5e,f. Specifically, most of the graphite surface with baseline electrolyte is covered with transition-metal species and “dead” Li (represented by $^7\text{Li}^-$ in Figure 5e and $^{58}\text{Ni}^-$ in Figure 5f). However, the coverage of these species is greatly reduced and the bulk graphite (represented by C_3^-) is readily exposed on graphite AEI with 1.5% LiBOB. Finally, concentrations of the decomposition products from carbonate solvents (represented by C_2HO^- in Figure 5d, as well as O_2^- , C_2HO^- and CHO_2^- in Figure S17, Supporting Information) and LiPF_6 (represented by PO^- , PO_2^- , POF_2^- and F_2^- in Figure S17, Supporting Information) are dramatically reduced with 1.5% LiBOB. Previous studies reported that the graphite AEI can be severely disrupted and reconstructed by the “communication” with the cathode side (e.g., transition-metal crossover),^[51,55] resulting in the continuous consumption of electrolyte. Since transition-metal ions only come from the cathode side, combining the TOF-SIMS analyses on both the cathode and anode sides, it is reasonable to correlate the carbonate solvents/ LiPF_6 decomposition on anode surface with the cathode surface stability, which means that AEI can be modified by tuning the cathode surface chemistry.

TOF-SIMS offers deeper insights into the configuration and microstructure of the graphite AEI, as shown in Figure 6. Figure 6a,b presents TOF-SIMS depth profiles of a series of secondary-ion fragments representing typical interphasial species. For graphite cycled with baseline electrolyte, the AEI displays a thick “three-layer” architecture: (i) deposited transition-metal species dissolved from the cathode and fluorine-enriched inorganic species (F_2^- in Figure 6a, $^{58}\text{NiF}_3^-$, CoF_3^- , $^6\text{LiF}_2^-$, and POF_2^- in Figure S18, Supporting Information), localized at the very surface of the AEI, (ii) penetrated transition-metal cations and oxygen-rich organic species from carbonate solvent decomposition (C_2HO^- , CHO_2^- , PO^- , O_2^- , and $^{58}\text{Ni}^-$ in Figure 6a, $^{18}\text{O}^-$, O_2H^- , CO_2^- , and PO_2^- in Figure S18, Supporting Information), dominating the intermediate AEI, and (iii) Li-enriched species, or “dead” Li ($^7\text{Li}^-$ in Figure 6a, $^6\text{Li}^-$, $^7\text{Li}_2^-$ in Figure S18, Supporting Information), comprising the interior layer of AEI. Note that the formation of fluorine-enriched species could be related to the high reactivity of exterior AEI with LiPF_6 or its decomposition species (e.g., PF_5 and HF). And the Li-enriched interior AEI arises from the kinetic barriers of Li-ion diffusion within the AEI, which leads to Li deposition during long-term cycling.^[56–59] On the other hand, for graphite cycled with 1.5% LiBOB (Figure 6b), the intermediate (oxygen-rich organic species) and exterior (dissolved transition-metal and fluorine-enriched inorganic species) layers merge and constitute a comparatively thinner “two-layer” AEI architecture with the interior (“dead” Li) layer, as opposed to the thick “three-layer” AEI for the baseline sample (Figure 6a). Such variations in the AEI architecture are also visualized in

Figure 6c–e with 3D distribution of representative fragments (F_2^- , O_2^- , and $^7\text{Li}^-$). As seen, both AEIs are composed of a F_2^- -containing exterior and a $^7\text{Li}^-$ -enriched interior layer. As for the O_2^- fragment, the majority is accumulated in the AEI intermediate region for the baseline sample, while for the 1.5% LiBOB sample, O_2^- emerges and dominates the near-surface region of the AEI. Additional detailed information is summarized in Figures S19 and S20 (Supporting Information). Such a straightforward comparison reconfirms that a thinner “two-layer” AEI is generated on the graphite surface brought by LiBOB in the electrolyte.

Based on the TOF-SIMS as well as the XPS analysis above, the synergistic effect of CEI chemistry on AEI with LiBOB incorporation in the electrolyte can be established: the superior electrochemical stability of boron-, oxygen-enriched CEI considerably alleviates parasitic electrolyte oxidation and transition-metal dissolution, thus significantly reducing chemical abuse (transition-metal ion attack) on AEI; meanwhile, the boron- and oxygen-enriched AEI on the anode side naturally possesses higher Li-ion diffusivity, which eliminates the kinetic barrier for Li-ion transfer (as evidenced by the improved rate capability of Li | graphite half cells in Figure S21, Supporting Information). As a result, AEI is free from the destruction–reformation processes, thus avoiding thickening and continuous consumption of active Li ions during regular battery operation. This mechanism is schematically illustrated in Figure 7. In the baseline electrolyte system, as cycling proceeds, transition-metal ions dissolve easily from the $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ cathode at high voltage and continuously migrate to the graphite anode surface. To compensate for the AEI damage caused by transition-metal ion attack, electrolyte decomposes continuously on anode surface. Ultimately, the AEI becomes substantially thick with a “three-layer” architecture accompanied by severe Li trapping and deteriorated Li-ion diffusivity (Figure 7a). Given the limited Li amount in a full cell, severe Li trapping depletes active Li ions from regular battery operation and significantly degrades the reversible capacity. In the meantime, the $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ cathode suffers from significant impedance buildup and voltage polarization due to undesired electrode–electrolyte reactions. However, with LiBOB, CEI is tuned with superior robustness that significantly suppresses the undesired electrode–electrolyte reactions during cycling (transition-metal dissolution and crossover in particular). As a result, the structural integrity of graphite AEI is well preserved with a favorable thin “two-layer” architecture (Figure 7b) and excellent electrochemical performance could be achieved.

3. Conclusion

By monitoring the EEI properties with LiBOB additive, the correlation between electrode surface chemistry and battery performance has been systematically investigated. The study reveals that B_xO_y species are introduced throughout the EEI by LiBOB sacrificial decomposition, which not only boosts the Li-ion diffusivity but also endows the charged cathode with superior electrochemical stability. Moreover, the physical architecture of the EEIs at the atomic level has been studied

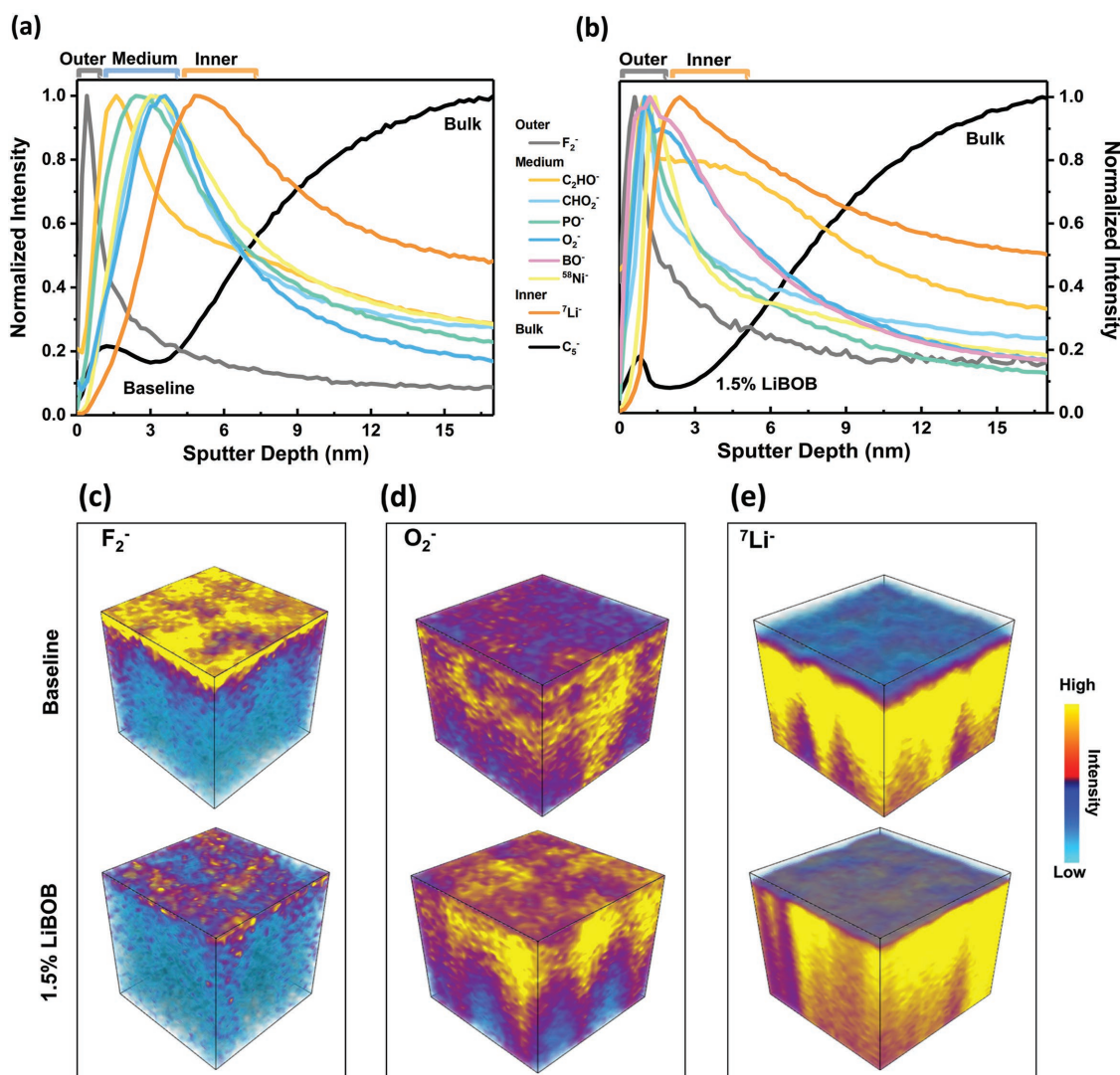


Figure 6. TOF-SIMS depth profiles (normalized to maximum) of fragments of interest acquired on graphite anodes retrieved from 500-cycle full cells with a) baseline electrolyte and b) 1.5% LiBOB. Corresponding 3D distributions of c) fluorine-enriched species (represented by F_2^-), d) oxygen-enriched species (represented by O_2^-), and e) trapped lithium (“dead” Li; represented by $^7Li^+$) in the graphite AEI with and without LiBOB in TOF-SIMS sputtered volume. Intensity of all fragments in the 3D view is normalized to maximum. Size of the TOF-SIMS sputtered volume is 100 μm (length) $\times$ 100 μm (width) $\times$ 17 nm (height).

with a surface-sensitive technique, TOF-SIMS. By enhancing the cathode surface stability (limiting transition-metal dissolution in particular), the AEI is reconstructed from a thick “three-layer” architecture to a “two-layer” architecture with O-enriched exterior layer and Li-enriched interior layer. Thus, the relationship between cathode surface chemistry and AEI configuration has been established. As a result of the electrode interphase modification, the service life of $LiNi_{0.94}Co_{0.06}O_2$ cathode is greatly extended via a manipulation of the electrode surface chemistry (capacity retention increases from 61 to 80% after 500 cycles in a full cell with a graphite anode). This work advances the understanding of EEI composition, configuration, and architecture as well as its influence on battery performance, which is critical for fabricating high-energy-density Li-ion batteries with long service life.

4. Experimental Section

Cathode Preparation: $Ni_{0.94}Co_{0.06}(OH)_2$ precursor ($\approx 12 \mu m$ in size) was synthesized by a transition-metal coprecipitation method. In a typical procedure, $NiSO_4 \cdot 6H_2O$ and $CoSO_4 \cdot 7H_2O$ with an appropriate molar ratio of Ni:Co = 94:6 were dissolved in distilled water, forming a combined concentration of 1.0 M in the mixture. Then, the mixed solution was pumped into a batch tank reactor (2.5 L), along with separate feeds of solution containing a proper amount of saturated KOH and NH_4OH . Experimental conditions like pH (≈ 11) and temperature ($\approx 50^\circ C$) were carefully controlled during the synthesis process. Finally, the coprecipitated powder was filtered, washed with distilled water, and then dried in the oven at $100^\circ C$ overnight. The as-synthesized $Ni_{0.94}Co_{0.06}(OH)_2$ precursor was then mixed with $LiOH \cdot H_2O$ with a proper molar ratio of Li:Ni + Co = 1:1.05, followed by 5 h of heating at $500^\circ C$ and 15 h of heating at $640^\circ C$ in oxygen atmosphere, yielding $LiNi_{0.94}Co_{0.06}O_2$. The morphologies of $Ni_{0.94}Co_{0.06}(OH)_2$ and

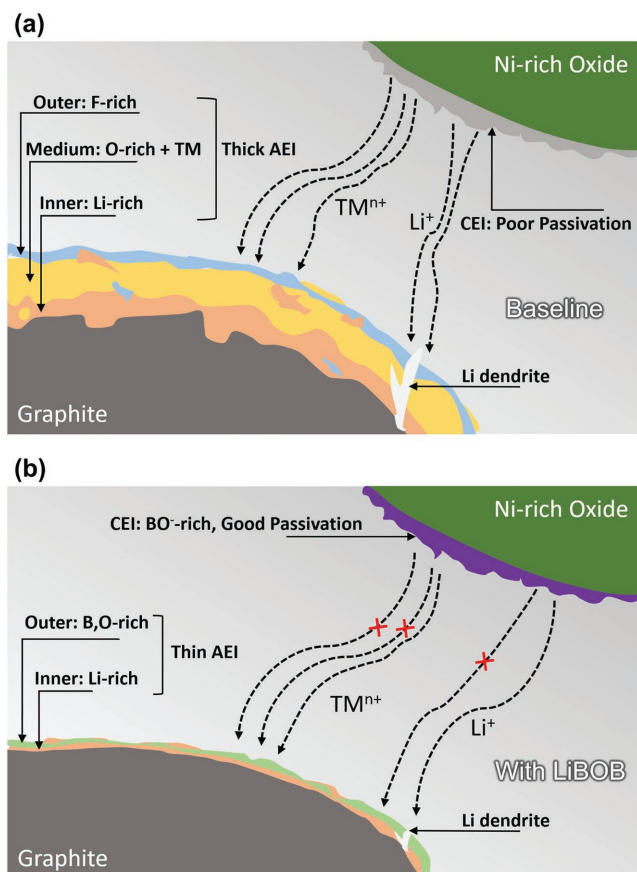


Figure 7. Schematic illustration of graphite AEL under influences of transition-metal crossover from Ni-rich layered oxide cathodes and lithium trapping during long-term cycling with a) baseline electrolyte and b) LiBOB-added electrolyte.

$\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ are shown in Figure S23 (Supporting Information), and the lithiated sample only suffers from a minor cation mixing issue (Figure S24 and Table S1, Supporting Information).

Electrolyte Preparation: The baseline electrolyte was composed of 1 M LiPF_6 in EC:DEC (1:1 by volume) solution from BASF Battery Materials and used as received. 0.3% LiBOB was prepared by adding 0.3 wt% LiBOB to the baseline electrolyte, followed by stirring for 2 d in argon-filled glove box before use. 1.5% LiBOB was prepared by the same way as the 0.3% LiBOB except for an increased LiBOB content of 1.5 wt%.

Material Characterization: XRD data were obtained with a Miniflex 600 X-ray diffractor at a scan rate of 1° min^{-1} . Structural morphology was characterized with a field-emission scanning electron microscope (SEM, Quanta 650). XPS measurement was carried out with an Axis Ultra DLD spectrometer (Kratos) with $\text{Al K}\alpha$ radiation. TOF-SIMS analysis was performed by a TOF.SIMS 5 spectrometer (ION-TOF GmbH). All detected secondary ions of interest had a mass resolution of >5000 and possessed negative polarity. A pulsed 30 keV Bi^+ (20 ns) ion beam set in a high current mode was employed for depth profiling and a 500 eV Cs^+ (negative) ion beam was utilized for the sputtering of the cycled electrodes with a typical sputtered area ($300 \mu\text{m} \times 300 \mu\text{m}$). The typical analyzed area was $100 \mu\text{m} \times 100 \mu\text{m}$. During XPS and TOF-SIMS measurements, all samples were transferred from an argon-filled glove box to the instruments with a novel air-free interface (U.S. Pattern Application Serial No. 14/445650). The sputter rate of TOF-SIMS was calibrated by averaging the sputtered depth of a continuous sputtering process of 44 h for cathode and 45 h for graphite anode with the corresponding working time. The sputter rate is calibrated as 0.04 nm s^{-1} for both the cathode and graphite anode.^[13,47]

Electrochemical Test: The cathode slurry was first prepared by mixing the active material ($\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$), Super P conductive carbon, and poly(vinylidene fluoride) (PVDF) (8:1:1 by weight) and dispersing them in N-methyl-2-pyrrolidone (NMP) under vigorous stirring. The obtained slurry was then cast on an aluminum foil with an active-material loading of $\approx 4 \text{ mg cm}^{-2}$. After drying, it was cut into round disks as the cathode. Half cells (CR2032 coin type) were assembled inside an argon-filled glove box using lithium metal as the anode. Long-term cycling tests were conducted with laminated pouch-type full cells consisting of $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ as the cathode and mesocarbon microbead (MCMB) graphite as the anode. Each pouch cell contained one anode ($4.83 \text{ cm} \times 8.15 \text{ cm}$) and one cathode ($4.78 \text{ cm} \times 8.15 \text{ cm}$), infused with around 0.2 g electrolyte. The N/P ratio was carefully controlled ≈ 1.1 . For full cells, two formation cycles were completed at C/10 rate, followed by cycling at C/3 rate at 4.3–2.5 V. A constant current–constant voltage (CCCV) mode for charging was applied to full cells, regardless of the difference in the electrolyte. For half cells, two formation cycles were conducted at C/10 rate, followed by cycling at different C rates at 4.4–2.8 V without no CCCV mode.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

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- [1] A. Manthiram, *J. Phys. Chem. Lett.* **2011**, 2, 176.
- [2] W. Li, B. Song, A. Manthiram, *Chem. Soc. Rev.* **2017**, 46, 3006.
- [3] A. Manthiram, J. Choi, *J. Power Sources* **2006**, 159, 249.
- [4] X. D. Zhang, J. L. Shi, J. Y. Liang, Y. X. Yin, J. N. Zhang, X. Q. Yu, Y. G. Guo, *Adv. Mater.* **2018**, 30, 1801751.
- [5] J.-L. Shi, D.-D. Xiao, M. Ge, X. Yu, Y. Chu, X. Huang, X.-D. Zhang, Y.-X. Yin, X.-Q. Yang, Y.-G. Guo, L. Gu, L.-J. Wan, *Adv. Mater.* **2018**, 30, 1705575.
- [6] V. Etacheri, R. Marom, R. Elazari, G. Salitra, D. Aurbach, *Energy Environ. Sci.* **2011**, 4, 3243.

- [7] J. M. Zheng, W. H. Kan, A. Manthiram, *ACS Appl. Mater. Interfaces* **2015**, 7, 6926.
- [8] F. Lin, I. M. Markus, D. Nordlund, T. C. Weng, M. D. Asta, H. L. Xin, M. M. Doeff, *Nat. Commun.* **2014**, 5, 3529.
- [9] Z. Moorhead-Rosenberg, A. Huq, J. B. Goodenough, A. Manthiram, *Chem. Mater.* **2015**, 27, 6934.
- [10] A. Manthiram, *ACS Cent. Sci.* **2017**, 3, 1063.
- [11] J. Y. Liang, X. X. Zeng, X. D. Zhang, P. F. Wang, J. Y. Ma, Y. X. Yin, X. W. Wu, Y. G. Guo, L. J. Wan, *J. Am. Chem. Soc.* **2018**, 140, 6767.
- [12] D. Aurbach, B. Markovsky, G. Salitra, E. Markevich, Y. Talyossef, M. Koltypin, L. Nazar, B. Ellis, D. Kovacheva, *J. Power Sources* **2007**, 165, 491.
- [13] W. Li, U. H. Kim, A. Dolocan, Y. K. Sun, A. Manthiram, *ACS Nano* **2017**, 11, 5853.
- [14] J. Lu, C. Zhan, T. Wu, J. Wen, Y. Lei, A. J. Kropf, H. Wu, D. J. Miller, J. W. Elam, Y. K. Sun, X. Qiu, K. Amine, *Nat. Commun.* **2014**, 5, 5693.
- [15] A. M. Wise, C. Ban, J. N. Weker, S. Misra, A. S. Cavanagh, Z. Wu, Z. Li, M. S. Whittingham, K. Xu, S. M. George, M. F. Toney, *Chem. Mater.* **2015**, 27, 6146.
- [16] J. Liu, A. Manthiram, *J. Electrochem. Soc.* **2009**, 156, A66.
- [17] P. Oh, S.-M. Oh, W. Li, S. Myeong, J. Cho, A. Manthiram, *Adv. Sci.* **2016**, 3, 1600184.
- [18] P. Yan, J. Zheng, M. Gu, J. Xiao, J. G. Zhang, C. M. Wang, *Nat. Commun.* **2017**, 8, 14101.
- [19] Y.-N. Zhou, J.-L. Yue, E. Hu, H. Li, L. Gu, K.-W. Nam, S.-M. Bak, X. Yu, J. Liu, J. Bai, E. Dooryhee, Z.-W. Fu, X.-Q. Yang, *Adv. Energy Mater.* **2016**, 6, 1600597.
- [20] A. J. Smith, J. C. Burns, S. Trussler, J. R. Dahn, *J. Electrochem. Soc.* **2010**, 157, A196.
- [21] S.-J. Tan, X.-X. Zeng, Q. Ma, X.-W. Wu, Y.-G. Guo, *Electrochem. Energy Rev.* **2018**, 1, 113.
- [22] S. Menkin, D. Golodnitsky, E. Peled, *Electrochem. Commun.* **2009**, 11, 1789.
- [23] J. H. Shim, R. Kostecki, T. Richardson, X. Song, K. A. Striebel, *J. Power Sources* **2002**, 112, 222.
- [24] M. Winter, *Z. Phys. Chem.* **2009**, 223, 1395.
- [25] M. Gauthier, T. J. Carney, A. Grimaud, L. Giordano, N. Pour, H. H. Chang, D. P. Fenning, S. F. Lux, O. Paschos, C. Bauer, F. Maglia, S. Lupart, P. Lamp, Y. Shao-Horn, *J. Phys. Chem. Lett.* **2015**, 6, 4653.
- [26] S. Y. Lang, R. J. Xiao, L. Gu, Y. G. Guo, R. Wen, L. J. Wan, *J. Am. Chem. Soc.* **2018**, 140, 8147.
- [27] N. P. W. Pieczonka, L. Yang, M. P. Balogh, B. R. Powell, K. Chemelewski, A. Manthiram, S. A. Krachkovskiy, G. R. Goward, M. Liu, J.-H. Kim, *J. Phys. Chem. C* **2013**, 117, 22603.
- [28] X. Wu, J. A. Shusterman, R. Marzke, C. A. Angell, *J. Electrochem. Soc.* **2004**, 151, A632.
- [29] D. Aurbach, Y. Talyossef, B. Markovsky, E. Markevich, E. Zinigrad, L. Asraf, J. S. Gnanaraj, H.-J. Kim, *Electrochim. Acta* **2004**, 50, 247.
- [30] M. Nie, B. L. Lucht, *J. Electrochem. Soc.* **2014**, 161, A1001.
- [31] C. Täubert, M. Fleischhammer, M. Wohlfahrt-Mehrens, U. Wietelmann, T. Buhrmester, *J. Electrochem. Soc.* **2010**, 157, A721.
- [32] J. W. Choi, D. Aurbach, *Nat. Rev. Mater.* **2016**, 1, 16013.
- [33] A. Manthiram, B. Song, W. Li, *Energy Storage Mater.* **2017**, 6, 125.
- [34] S. Xin, Y. Guo, L. Wan, *Sci. Sin. Chim.* **2011**, 41, 1229.
- [35] J. Jiang, J. R. Dahn, *Electrochem. Solid State Lett.* **2003**, 6, A180.
- [36] K. Xu, U. Lee, S. Zhang, M. Wood, T. R. Jow, *Electrochem. Solid State Lett.* **2003**, 6, A144.
- [37] C. Liao, K. S. Han, L. Baggetto, D. A. Hillesheim, R. Custelcean, E.-S. Lee, B. Guo, Z. Bi, D. E. Jiang, G. M. Veith, E. W. Hagaman, G. M. Brown, C. Bridges, M. P. Paranthaman, A. Manthiram, S. Dai, X.-G. Sun, *Adv. Energy Mater.* **2014**, 4, 1301368.
- [38] T. Ohzuku, A. Ueda, M. Nagayama, *J. Electrochem. Soc.* **1993**, 140, 1862.
- [39] W. Li, J. N. Reimers, J. R. Dahn, *Solid State Ionics* **1993**, 67, 123.
- [40] S.-K. Jung, H. Gwon, J. Hong, K.-Y. Park, D.-H. Seo, H. Kim, J. Hyun, W. Yang, K. Kang, *Adv. Energy Mater.* **2014**, 4, 1300787.
- [41] J. Li, W. Li, S. Wang, K. Jarvis, J. Yang, A. Manthiram, *Chem. Mater.* **2018**, 30, 3101.
- [42] J.-G. Han, J. B. Lee, A. Cha, T. K. Lee, W. Cho, S. Chae, S. J. Kang, S. K. Kwak, J. Cho, S. Y. Hong, N.-S. Choi, *Energy Environ. Sci.* **2018**, 11, 1552.
- [43] A. Rezzita, M. Sauer, A. Foelske, H. Kronberger, A. Trifonova, *Electrochim. Acta* **2017**, 247, 600.
- [44] R. Dedryvere, L. Gireaud, S. Grugeon, S. Laruelle, J. M. Tarascon, D. Gonbeau, *J. Phys. Chem. B* **2005**, 109, 15868.
- [45] X. Li, J. Zheng, M. H. Engelhard, D. Mei, Q. Li, S. Jiao, N. Liu, W. Zhao, J. G. Zhang, W. Xu, *ACS Appl. Mater. Interfaces* **2018**, 10, 2469.
- [46] Q. Li, H. Chou, J. H. Zhong, J. Y. Liu, A. Dolocan, J. Zhang, Y. Zhou, R. S. Ruoff, S. Chen, W. Cai, *Nano Lett.* **2013**, 13, 486.
- [47] W. Li, A. Dolocan, P. Oh, H. Celio, S. Park, J. Cho, A. Manthiram, *Nat. Commun.* **2017**, 8, 14589.
- [48] M. R. Busche, T. Drossel, T. Leichtweiss, D. A. Weber, M. Falk, M. Schneider, M. L. Reich, H. Sommer, P. Adelhelm, J. Janek, *Nat. Chem.* **2016**, 8, 426.
- [49] W. Li, X. Liu, H. Celio, P. Smith, A. Dolocan, M. Chi, A. Manthiram, *Adv. Energy Mater.* **2018**, 8, 1703154.
- [50] S. Nowak, M. Winter, *Acc. Chem. Res.* **2018**, 51, 265.
- [51] J. Kim, H. Ma, H. Cha, H. Lee, J. Sung, M. Seo, P. Oh, M. Park, J. Cho, *Energy Environ. Sci.* **2018**, 11, 1449.
- [52] Y. Yao, *Natl. Sci. Rev.* **2017**, 4, 17.
- [53] N. W. Li, Y. Shi, Y. X. Yin, X. X. Zeng, J. Y. Li, C. J. Li, L. J. Wan, R. Wen, Y. G. Guo, *Angew. Chem., Int. Ed.* **2018**, 57, 1505.
- [54] X.-X. Zeng, Y.-X. Yin, Y. Shi, X.-D. Zhang, H.-R. Yao, R. Wen, X.-W. Wu, Y.-G. Guo, *Chem* **2018**, 4, 298.
- [55] C. Zhan, J. Lu, A. J. Kropf, T. Wu, A. N. Jansen, Y. K. Sun, X. Qiu, K. Amine, *Nat. Commun.* **2013**, 4, 2437.
- [56] K. Xu, A. von Cresce, U. Lee, *Langmuir* **2010**, 26, 11538.
- [57] Y. Yamada, Y. Iriyama, T. Abe, Z. Ogumi, *Langmuir* **2009**, 25, 12766.
- [58] J. Chen, C. Wu, C. Tang, W. Zhao, M. Xu, C. M. Li, *ChemistrySelect* **2017**, 2, 5518.
- [59] Z. Jian, M. Zheng, Y. Liang, X. Zhang, S. Gheyhani, Y. Lan, Y. Shi, Y. Yao, *Chem. Commun.* **2015**, 51, 229.